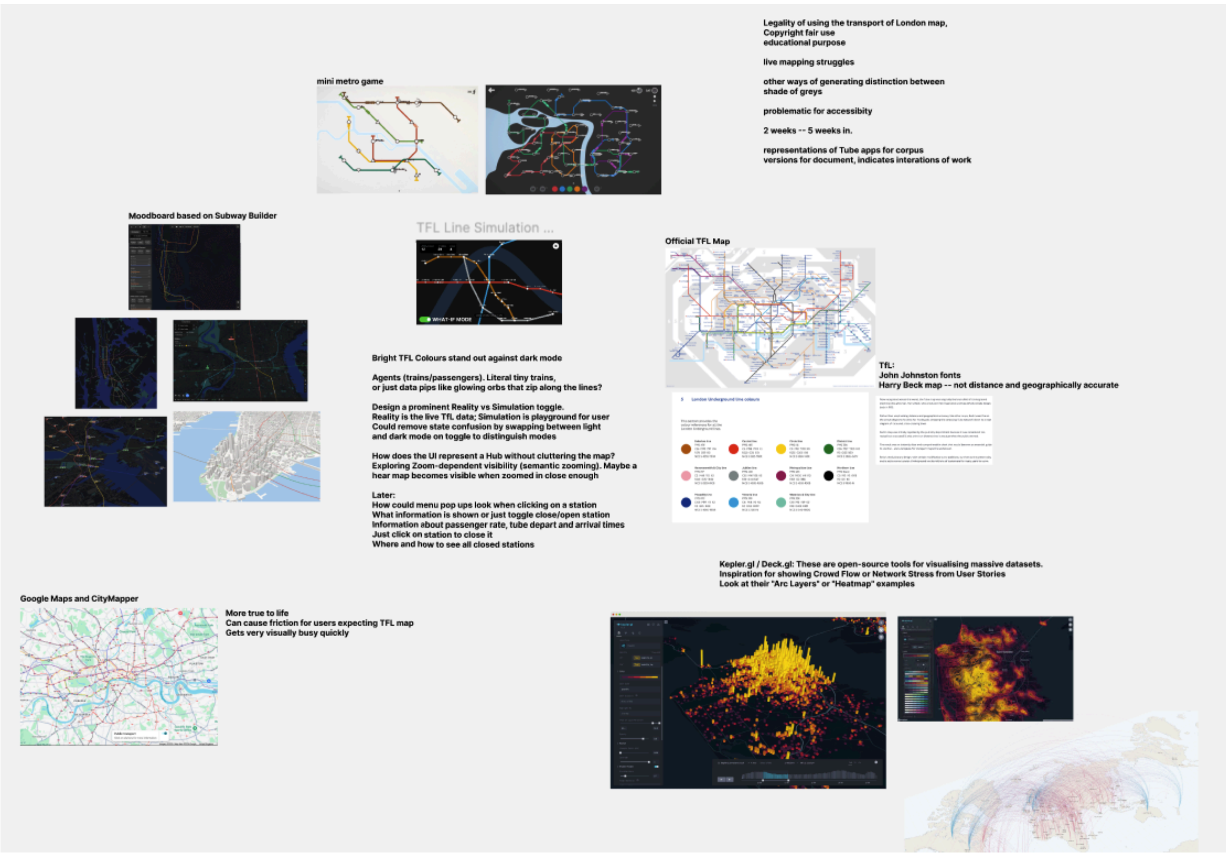
DESIGN RESEARCH

London Underground
What-If Simulator

Visual moodboard - UX research - Brand and colour theory

This document traces the design thinking behind the simulator - from the original Figma moodboard through to the production interface. It covers visual inspiration, UX research, brand heritage, colour theory and typographic rationale.

Figma moodboard: https://figma.com/file/nuw9BCsmE4yIkZqhpWC6zn



1. Original Figma Moodboard

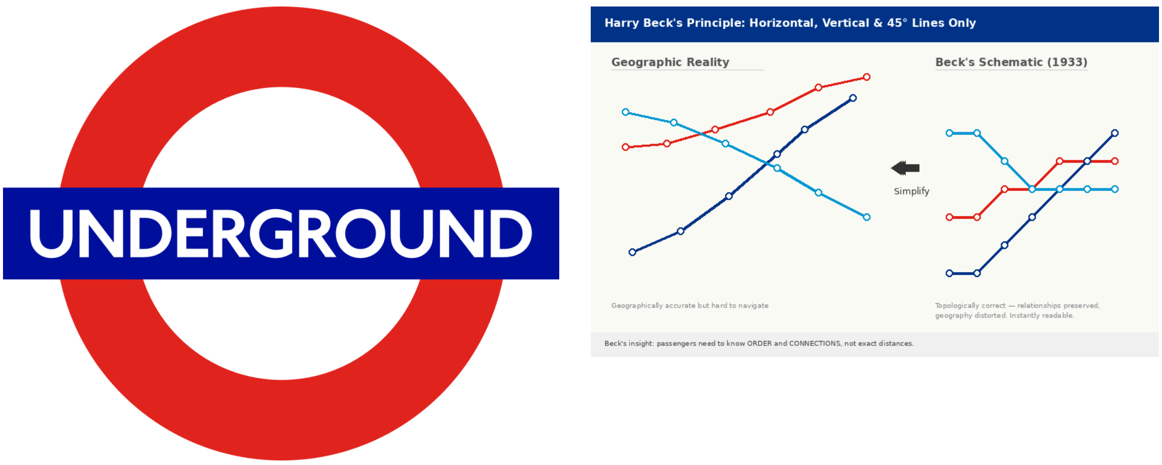
The project began with a Figma moodboard that set the directional aesthetic before any code was written. The board established three pillars that would guide all subsequent design decisions:

- Dark, immersive interface - map as the hero, UI floating above it.
- TfL brand language - official line colours, transport typography heritage.
- Data-dense but readable - mission control aesthetic, not consumer-app simplicity.

2. TfL Brand Heritage

The London Underground has one of the most recognisable and rigorously maintained visual identities in the world. Understanding this heritage is essential because the simulator borrows directly from it - using official line colours, acknowledging the tube map's topological conventions and working within the cultural expectations of a London audience.

2.1 The Roundel

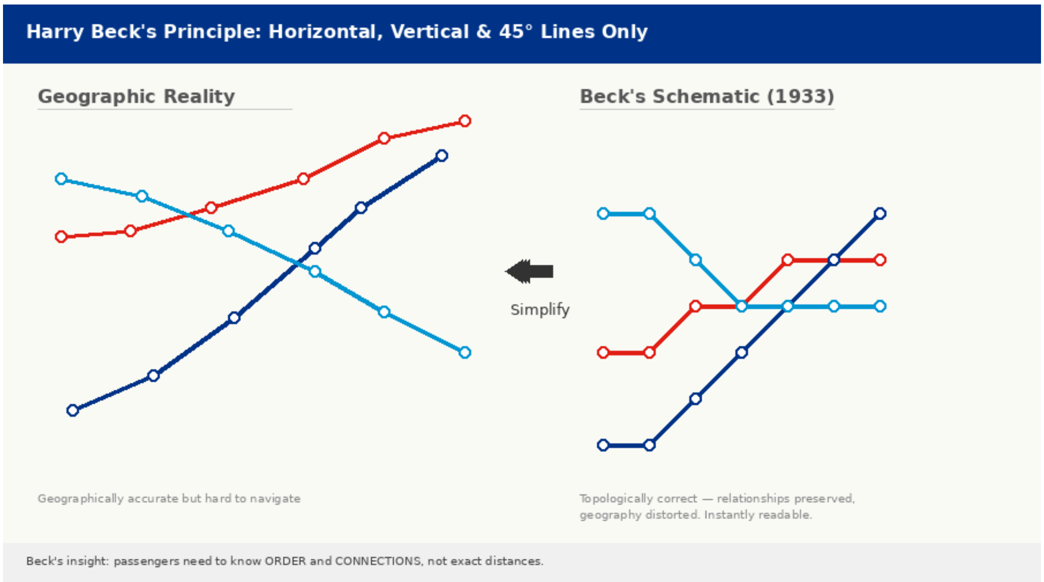


Left: TfL roundel (bar-and-circle, 1908–present). Right: Harry Beck's core principle - schematic topology over geographic accuracy.

The TfL roundel (bar-and-circle) was first used at stations in 1908. Its power comes from its simplicity: a red ring and blue bar that frame station names against chaotic station environments and adverisments. The design has been standardised and trademarked and is now used across all TfL services. It is voted one of Britain's most recognised trademarks.

The simulator's title treatment echoes this as the words 'What-if Simulator' are always rendered in the current accent colour (blue or amber) while the rest of the title is muted. A brand highlight within a neutral frame.

2.2 Harry Beck's Tube Map (1933)



Beck's radical insight: show connections and order, not geographic distance. Only horizontal, vertical, and 45° lines.

Harry Beck's 1931 design (published 1933) is considered one of the greatest works of information design ever produced. As an engineering draftsman, Beck applied circuit-diagram logic to the tube map: straighten lines to horizontal, vertical, and 45° diagonals; compress outer zones; expand the centre.

This has direct implications for the simulator's design:

- The map layer uses Leaflet.js with geographic accuracy (real lat/long), but the visual style deliberately echoes Beck's aesthetic: bright coloured lines on a dark background, minimal geographic labelling, emphasis on connections.
- Station dots scale with zoom level (7px at z12 → 15px at z16), prioritising readability over precision, just as Beck did.
- The route highlighting draws a thicker polyline on top of existing lines, communicating 'this is your path' without needing a separate diagram.

Research Note	Beck's map is a topological diagram, not a geographic one. The simulator occupies an interesting middle ground: it uses real geography (Leaflet/OpenStreetMap) but styles it to look like Beck's schematic.

2.3 Johnston Typeface: The Original TfL Font

In 1913, Edward Johnston was commissioned to design a new typeface exclusively for London Underground. Johnston Sans became the typeface of the London Underground and remained so for over a century, with a digital revival (Johnston 100) in 2016.

Johnston's design principles (geometric simplicity, high legibility, humanist warmth) directly influenced the simulator's typeface choice:

Property	Johnston Sans (TfL)	Geist (Simulator)
Origin	London Underground, 1916	Vercel / Basement Studio, 2023
Classification	Humanist sans-serif	Geometric/humanist sans-serif
x-height	Moderate	High - optimised for screens
Weight range	Regular–Bold	Thin–ExtraBold (9 weights)
Context	Signage, print, wayfinding	Digital UI, developer tools
Legibility	Excellent at large sizes	Excellent at small screen sizes

Geist was chosen as a conscious modern counterpart to Johnston as both are clean, functional sans-serifs optimised for their medium, and both carry a sense of technical precision without coldness.

3. Colour Research and Theory

3.1 TfL Official Line Colours

All 11 TfL line colours as used in the simulator, official Transport for London palette.

The line colours are not arbitrary; each has decades of cultural association in London. Passengers do not read station names first; they identify lines by colour. The simulator uses the exact official hex values, sourced from TfL's corporate design standards, to maintain this legibility.

Key colour challenges on a dark background:

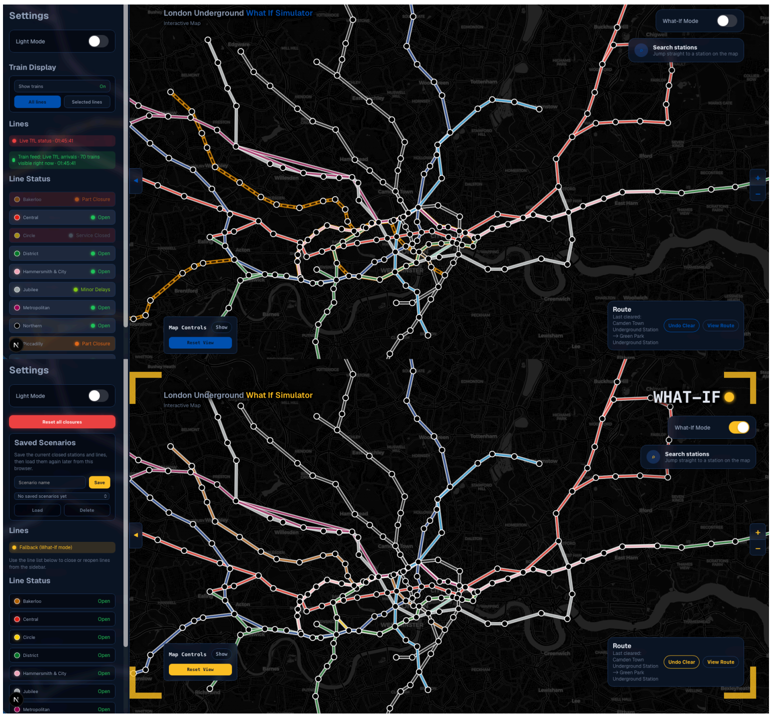
- Northern line (#000000, black) - invisible on a black map. Solved with an 8px white outline behind the 4px black stroke.
- Jubilee (#A0A5A9, silver-grey) - low contrast on dark. Kept legible via the white outline treatment shared with Northern.

3.2 UI Palette - Dark Theme

#0A0F1A	**Background**	Deep navy/black map bg
#0F172A	**Card / Panel**	Frosted panel bg
#1E3A5F	**Border**	Panel borders
#94A3B8	**Text**	Primary labels
#64748B	**Text Muted**	Subtitles & timestamps
#E2E8F0	**Text Light**	High-emphasis values
#3B82F6	**Accent Blue**	Normal-mode accent
#FBBF24	**Accent Amber**	What-If mode accent
#22C55E	**Success**	Live / open status
#F59E0B	**Warning**	Fallback / partial
#EF4444	**Danger**	Closed / error

The complete UI palette - all colours drawn from Tailwind's slate/blue scale, expressed as CSS custom properties.

3.3 Accent Colour Duality



Blue (#3B82F6) in normal mode; amber (#FBBF24) in What-If mode. Every interactive element shifts.

The single most distinctive colour decision in the simulator is the accent shift between modes. This was not in the original moodboard; it evolved during development to solve a real UX problem to clarify whether station clicks would plan a route or close the station.

The solution draws from two research traditions:

Alert / Mode Signalling

Industrial control systems (nuclear plants, air traffic control, military systems) use colour-coded operational modes. Red/amber modes signal heightened states where actions have greater consequences. The simulator borrows this metaphor; amber is traditionally a 'caution' colour, appropriate for a hypothetical simulation mode where actions are reversible but significant.

Ambient Awareness

Rather than a modal dialogue or overlay that blocks interaction, the mode shift uses ambient colour change, every interactive element changes colour simultaneously. This creates what interaction designers call ambient awareness: the state is always peripherally visible without demanding focused attention.

Design Principle	Users should never have to think about which mode they're in. The amber accent makes the current mode legible at a glance, without any explicit mode indicator label needing to be read.

4. UX Research and Design Patterns

4.1 Glass-morphism

Glass-morphism (frosted glass UI) is the primary panel language of the simulator. Every floating UI element (sidebar, search box, route panel, HUD) uses the same recipe:

Property	Value and Rationale
background	rgba(15, 23, 42, 0.80) - 80% opacity dark navy. Transparent enough to show the map context underneath.
backdrop-filter	blur(8–12px) - blurs the map behind the panel, creating visual separation without a solid background.
border	1px solid #1E3A5F - a subtle blue-grey line that defines the panel edge against both the dark background and the slightly lighter blurred map.
box-shadow	0 12–16px 36–42px rgba(0,0,0,0.28–0.45) - a deep shadow grounds the panel above the map.

This treatment is a deliberate design choice over solid panels: the user always maintains spatial awareness of the map below. They can see train positions, line layouts and their route highlighted even while interacting with controls.

4.2 Search Box UX - Station Search Panel

The station search panel evolves through three states, each revealing more context:

1. **Collapsed pill button -** shows search icon + 'Search stations'. Minimum footprint on screen.
2. **Expanded search panel -** 320px panel with text input and results list. Results highlight on hover/keyboard navigation.

3. **Station action card -** appears below results when a station is selected. Shows current route context (start/destination chips) and contextual action buttons. Changes content entirely in What-If Mode.

This progressive disclosure pattern, the UI reveals complexity only when needed, is drawn from the Information Architecture principle of 'Layers of complexity': show the minimal interface by default, expand on demand.

4.3 Layout Architecture with a Map-First approach

The layout is fundamentally different from a traditional web application: there are no pages, no navigation bars, no content areas. The entire application is a single full-viewport canvas with floating control panels.

This approach is grounded in research on spatial cognition and map-based interfaces:

- Continuity - the map never moves or resizes when controls are opened. Users maintain their spatial orientation.
- Context preservation - the sidebar slides over the map rather than pushing it. The user can see through the frosted glass panel to maintain spatial awareness.
- Minimal interruption - controls appear where they are needed (search top-right, route panel bottom-right, HUD bottom-left) following the 'corners' convention from game HUD design.

4.4 What-If Mode

The What-If Mode is the simulator's defining feature and the most complex UX challenge: how do you clearly communicate to a user that their clicks now have different consequences?

The design uses three simultaneous signals:

Signal	Implementation	Cognitive role
Colour shift	Blue → Amber on all interactive elements	Ambient/peripheral awareness - no reading required
Corner brackets	Four 80×80px amber L-brackets at viewport corners	Spatial frame - 'you are inside the simulation'
WHAT-IF label	40px Geist Mono, top-right, pulsing dot	Explicit mode label for any doubt

The corner bracket motif is drawn from film and photography - viewfinder framing brackets signal 'you are looking through a lens at a constructed view'. Applied to the simulator, it communicates: this is not the real network, it is a view of a hypothetical one.

4.5 Mobile UX and adapting the interface

Mobile portrait presents a fundamentally different spatial problem: the viewport is narrow and tall, the touch target requirements are larger and hover interactions don't exist. The simulator ships a completely separate layout for mobile portrait, derived from the same components:

Component	Desktop	Mobile Portrait
Controls	Slide-in left sidebar	Bottom sheet (slides up from bottom edge)
Search	Top-right collapsible panel	Always-visible full-width bar below title
Route panel	Fixed bottom-right, 340px	Full-width floating card, 52vh max height
Mode toggle	Sidebar toggle (What-If switch)	Persistent bottom bar in What-If Mode
Delay expand	Hover (750ms debounce)	Tap to toggle

The bottom sheet follows the iOS/Android design convention for progressive disclosure on mobile - content that needs more space slides up from the bottom rather than appearing as a modal. The route panel uses 52vh maximum height to always leave at least 48% of the viewport as interactive map.

5. Competitor and Reference Analysis

The simulator was benchmarked against three categories of reference products: official transit apps, third-party navigation apps, and data visualisation tools.

5.1 TfL Official App and Website

Dimension	TfL Official	Our Simulator
Background	White/light theme	Dark, map-first
Map style	Standard street map	Schematic dark-map, tube lines dominant
Line colours	TfL official palette	Same palette - direct continuity
Simulation	None - live only	Core feature - hypothetical closures
Train positions	Journey planner only	Live animated trains on map
Differentiator	Official, authoritative	Exploratory, visual, immersive

5.2 Citymapper

Citymapper is the best-known third-party London transit app. Key design observations:

- Citymapper uses a white/light default theme. Notably, it has no official dark mode - when phone dark mode is applied, the map (being a raster image) stays white while the UI flips, creating visual inconsistency. The simulator avoids this entirely by being dark-first throughout.

- Citymapper prioritises multi-modal routing (Tube + Bus + Walk). The simulator focuses exclusively on the Underground - a narrower scope enables a deeper, more specialised interface.
- Citymapper uses bold colour blocks for UI elements. The simulator uses the line colours only on the map and status indicators, never as background fills for panels, maintaining the map's visual dominance.

6. Design Principles Summary

The research above crystallises into six core principles that should guide any design decisions on this project:

1. Map First	The map is always the primary content. No UI element should permanently obscure it. Panels float, blur through to the map, and slide away.
2. Colour Carries Meaning	Every colour choice is functional, not decorative. Line colours identify lines. Green = live/open. Amber = caution/simulation. Red = closed/error. Introducing new colours for decorative purposes would erode this system.
3. Mode Should Be Unmistakable	Users must never be uncertain which mode they are in. Ambient colour shift + spatial framing (corner brackets) solve this without interrupting interaction.
4. Progressive Disclosure	Show the minimal interface first. The search box is small until it is interacted with. The route panel is a summary bar until expanded. The sidebar is off-screen until toggled.
5. Context Preservation	Users should never lose their place on the map. Panels slide over (not replace) the map. Frosted glass keeps the map partially visible at all times.
6. Platform Conventions	Desktop and mobile are treated as distinct platforms with distinct conventions - hover vs. tap, sidebar vs. bottom sheet, click-to-set vs. contextual action card.

Sources and references

- Transport for London - The Roundel: tfl.gov.uk/corporate/about-tfl/culture-and-heritage/art-and-design/the-roundel
- London Transport Museum - Transforming the Tube map: ltmuseum.co.uk/collections/stories/design/transforming-tube-map-harry-becks-iconic-design
- Basement Studio - The Birth of Geist: https://basement.studio/post/the-birth-of-geist-a-typeface-crafted-for-the-web
- Vercel - Geist Font & Design System: vercel.com/font
- Creative Review - Edward Johnston's London Underground logo: creativereview.co.uk/london-underground-logo-edward-johnston

- Original Figma Moodboard: figma.com/file/nuw9BCsmE4yIkZqhpWC6zn (TFL What-If, Jan 2026)
- Anglotees - How the London Tube Map Distorts Geography: anglotees.com/how-the-london-tube-map-distorts-geography-and-why-it-works-anyway